Wireless Personal Communications
https://doi.org/10.1007/s11277-021-08712-9

Mutual Information Feature Selection (MIFS) Based Crop Yield Prediction on Corn and Soybean Crops Using Multilayer Stacked Ensemble Regression (MSER)

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Accepted: 23 June 2021
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Abstract
Crop yield prediction model helps the farmers in order to make better decisions about the appropriate time to cultivate the crops and what types of crops to be cultivated based on environmental factors to produce better yield. Advanced ensemble regression crop yield prediction model is to predict the crop yield based on the phenotype factors includes precipitation, solar radiation, maximum temperature, minimum temperature etc. The corn and soybean crops dataset includes 38 years of yield performance data collected across 105 different locations. Comparative analysis made between correlation and mutual information on the basis of feature selection. Most related features towards crop yield and achieved better predictions on crop yield though mutual information. Our proposed mutual information based advanced ensemble regression technique involved in the prediction process of crop yield on corn and soybean crops and achieved good prediction accuracy based on phenotype factors. The predicted yield performance of advanced ensemble regression crop prediction model outperformed several supervised machine learning and advanced ensemble learning algorithms. Various regression accuracy parameter metrics such as mean absolute error, mean square error and root mean square error are also involved in performance measures. Our prediction results also ensure that weather and crop management parameters are most influential towards crop yield prediction rather than soil parameters.

Keywords Multiple linear regression · Decision tree regression · K-nearest neighbour support vector machine · Random forest · Gradient boosting regression · Ensemble regression

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Published online: 30 June 2021

1 Introduction

Many of the countries across the world are still facing the problem of huge food supply chain management demand due to rapid industrialisation and increasing population. Still farmers across the globe were not able to get adequate yield of their cultivated crops due to drastic climatic changes, lack of water sources for irrigation management, lack of crop nutrients, inadequate rainfall and severity of crop disease. Lack of knowledge about the prediction on crop cultivation in order to predict the correct possibilities of the crop to produce good yield or not. If proper prediction on crop cultivation or crop yield prediction was not done, it would drastically affect the farmers time period, work effort and financial investment on crops. Crop yield prediction model helps the farmers in order to make better decisions about the appropriate time to cultivate the crops and what types of crops to be cultivated based on environmental factors to produce better yield. The proposed work focused on prediction on crop yield based on the parameters like climatic, Edaphic and crop management factors.Applying these Parameters with advanced ensemble machine learning comprises appropriate training and testing provide the right direction on prediction about the crop yield for the upcoming years.

Nowadays, human work and time can be significantly reduced due to technological development all over the world. In the Agricultural sector also, there are a number of recent technologies involved for betterment of crop production. Internet of things (IoT) [1], provides promising technology in the field of precision agriculture by fixing various sensors, actuators, cameras etc., in the agricultural land to monitor the crop growth, water level, temperature, disease identification, soil moisture etc., It results automation in crop management such as irrigation, disease and fertilizers. Cloud [2], acting as the repository of crop related data were collected from various IoT devices fixed in the agricultural field. Crop repository can be remotely accessible by the experts and perform useful computation for betterment of crop production. Data mining [3], is the process of extracting relevant crop related data from the repository, then making useful decisions on crop management practices and different data mining algorithms were involved in discovery of hidden patterns for the increased crop productions. Artificial intelligence and machine learning [4], rapidly created a significant impact in the field of precision agriculture. Many supervised and unsupervised machine learning are involved in parsing the crop data, learning from it and making decisions without human intervention. It significantly plays a vital role in crop production forecasting, crop classification, weed detection, crop disease classification and prediction etc., Image processing [5–7], gives the significant breakthrough in identification and classification of crop disease in the earlier stages of crop growth which adversely affected the crop production. Machine learning image classifiers are used to classify the diseased crops from the healthy crops. Remote sensing [8], also engaged and plays a predominant role in assessment of crop damage, forecasting crop production, weeds detection, disease identification, assessment of crop condition, monitoring of irrigation etc., Unmanned aerial vehicles (UAVs) and Drones are widely used for crop monitoring with high resolution and low cost. Several recent technologies [9], also provide support to strengthening the precision farming named as automated steering system, geo-mapping, variable rate technology, and global positioning system etc., used in various crop management practices such as crop disease, irrigation, pesticides and crop production forecast.

The organisation of the paper is defined as follows: Sect. 2 describes about the related works of crop yield prediction applying machine learning, Sects. 3 and 4 mentioned about data availability and feature selection, Sect. 5 elaborates models analysed for the

identification of weak learners in proposed work, Sect. 6 discussed about methodology used for proposed work, Sect. 7 deals about result and discussions of proposed work, Sect. 8 discussed about conclusion.

2 Related Works

Convolutional Neural Network (CNN) [10], used in crop production estimation covered moderate resolution imaging spectroradiometer (MODIS) data and weather data for the duration of 14 years. CNN outperformed artificial neural network (ANN), stacked-sparse auto encoder (SSAE) with RMSE as an accuracy metrics. Support vector machine (SVM) [11], engaged in maize crop yield estimation and classification with data obtained from remote sensing and suitable crop modelling involved. SVM produced higher prediction accuracies with other machine learning algorithms such as decision tree (DT) and random forest (RF) in terms of less RMSE and MAE values in prediction as well as classification. RF [12], also engaged in the process of sugarcane crop yield estimation with feature selection technique as forward feature selection and for achieving higher accuracy prediction in yield, hyper tuning parameter used lies in the baselines of yield prediction by the human experts as well as the measures of previous year crop yield.

SVM and RF [13], employed in yield prediction of wheat crop, used integrated climatic assessment indicator (ICAI) for statistical modelling including meteorological parameters. RF outperformed SVM in terms of prediction accuracy along with regression accuracy parameter metrics such as RMSE, MAE and R^2. [14], analysed all the predictive ability of the machine learning regression algorithms for the estimation of grain crop production. Data set used for this predictive analysis was collected from multi-farm based on the different crop sowing seasons such as pre, mid and late seasons. Considering all the observations RF crop yield predictions exceeded other predictive machine learning algorithms. [15], analysed the impact of soil factors affecting the coffee crop yield. Soil organic components such as nitrogen, pH, zinc, magnesium etc., are taken into the observations or independent variables of crop yield estimation. Extreme machine learning (EML) model used for the coffee crop yield estimation and it outperformed MLR and FR in terms of higher range of prediction with suitable accuracy parameter metrics such as MAE and RMSE.

Hierarchical machine learning [16], results in the estimation of single crop production on the various varieties of seeds. Most influential weather parameters towards better crop yield production of different seeds varieties was done through suitable weather forecast models. Integration of weather and crop production forecast models predicted optimal weather parameters and yield values of respective seeds categories. Hierarchical machine learning produced better yield prediction accuracy than RF and linear regression (LR). Boosted tree classifiers (BTC) and Gradient Boosted Regression Trees (GBRT) [17] used for the prediction of weekly irrigation plan with respect to various categories of crop. Irrigation data set used for this weekly irrigation recommendation plan consists of data collected from various sensors placed in the agricultural field with the duration period of two years. BTC and GBRT outperformed other classification and regression machine learning algorithms in terms of prediction and classification accuracy. [18], RF includes 30 years of crop data consisting of weather parameters and genotype factors across the counties of the United States were taken into the considerations of yield prediction. By fixing the yield prediction accuracy of MLR as a benchmark, RF performed well in terms of accuracy with respect to error metrics RMSE. Supervised Kohonen Networks (SKNs) [19], results in the

yield estimation of wheat crop based on the satellite images of various kinds of soil layers spread across the agricultural fields and different growth level images of crop were taken as the predictor or independent variables of crop yield measurement or estimation. Compared to ANN models, (SKNs) produced higher accuracy in yield prediction based on satellite images.

Spatial NN and Temporal NN networks are involved in crop yield prediction considering major spatial and temporal data affecting the crop yield. Spatial NN produced higher accuracy rate [20]. ANN used for sugarcane crop yield prediction based on lignocellulose biomass in terms of one of the accuracy parameters R^2 [21]. Deep learning involved in prediction the crop yield based on the Convolutional Architecture for Fast Feature Embedding (Caffe), which is an estimation model of deep learning [22]. Both Statistical and Neural methods are used for crop yield predictions. Statistical methods such as MLR, Projection pursuit regression (PPR) are producing lower accuracy compared to feed-Forward neural network model [23]. ANN used effectively for rice crop yield prediction in the northern region of Iran, this study also found potential rice crop production majorly depends on pH value of soil, organic content in the soil and electrical conductivity [24, 25]. Artificial neural networks-gray wolf optimizer (ANN-GWO), which is a hybrid machine learning algorithm introduced for effective crop yield prediction. It provided better accuracy values in yield prediction compared to artificial neural networks-imperialist competitive algorithms (ANN-ICA) related with metrics such as R, RMSE and MEA [26, 27]. Six machine learning algorithms such as Lasso, SVM, RF, XGBoost, LSTM and CNN used for wheat crop yield prediction based on environmental variables. XGBoost provided better accuracy values compared to other ML algorithms [28].

From the understanding of the related works, several issues are still acting as the challenging factors of effective crop yield prediction or estimation, they are insufficient predictor variables in terms of weather, soil and other management factors. Even though various supervised machine learning and ensemble algorithms are involved in the process of crop yield prediction, it lacks effective feature selection techniques and heterogeneous approach for better prediction. Our proposed crop yield prediction model had covered most of the sufficient predictor variables with effective feature selection techniques and heterogeneous approach used for crop production estimation.

3 Data Availability and Dimensions

Corn and Soybean crop yield performance dataset collected between the years 1980 and 2018 across 105 different corn and soybean belt locations covering 13 states of the USA have been used in this crop yield prediction model [29]. Corn and soybean dataset includes four categories of data namely yield observations, environmental data, soil data, and crop management data. The data analysed for the prediction of corn crop yield only includes phenotype factors. No Genotype factors are considered for this proposed work. In [30], results of study revealed that prediction on crop yield based on environmental factors i.e. phenotype had more impact than biological factors i.e. genotype.

Yield performance and crop management data analysed in this work were collected from USDA-National Agricultural Statistics Service [31]. Environmental or Weather data acquired from Daymet: Daily Surface Weather Data on a 1- km Grid for North America, Version 3 [32]. Soil data from Gridded Soil Survey Geographic Database for the United States (gSSURGO) [33].

Considering the dimension of the dataset, yield measurement data considered as dimension of output data. Dimensions of input data can be divided into a number of components with respect to weather, soil, and management data.

- Weather data consist of six components namely precipitation, solar radiation, snow water equivalent, maximum temperature, minimum temperature and vapour pressure were represented in the csv file as W_{ab}, where 'a' is the index of weather component and 'b' is the index of week of year. a = 1, 0.6, b = 1, … ,52.
- Soil data consist of the following components

1. 10 components namely wet soil bulk density, dry bulk density, clay percentage, upper limit of plant available water content, lower limit of plant available water content, hydraulic conductivity, organic matter percentage, pH, sand percentage and saturated volumetric water content variables were observed at different depth can be represented as S_{ab}, where 'a' is the index of soil component and 'b' is the index of depth of soil. a = 1, 0.10, b = 1, 0.10.
2. 4 components namely field slope, productivity index, average Productivity index and root depth observed at the surface can be represented as SS_a where 'a' is the index of surface of soil.

Crop management data include planting dates consisting of 16 components in terms of week and can be represented as P_a, where a is the index of planting date week.

4 Feature Selection on Corn and Soybean

Proper feature engineering and feature selection process are acted as the backbone for any machine learning model to produce higher accuracy of results in prediction and classification. Initially the dataset of corn and soybean crop consisting of 434 and 432 phenotype features were taken into consideration for crop yield prediction. The phenotype features includes weather, soil and management practices data with respect to various locations and years. Table 1 describes the total number of variables considered for each phenotype feature of corn and soybean crop.

4.1 Missing Values

Initially there are some missing values available in corn and soybean crop dataset. There are around 7% of missing values in soil data for some regions which can be replaced by mean values of the same soil variable of other counties. Management data also has around

Table 1 Measure of phenotype features of corn and soybean crop

Name of the crop	Period of years	Covered locations	Total number of parameters			Overall parameters
			Weather	Soil	Management	
Corn	38	105	312	104	16	434
Soybean	38	105	312	104	14	432

6% of missing values which can be substituted by the mean value of the same management variable at the same year of other counties. Median and mode method were tried for replacement of missed values but mean method produced higher predictions. There are no missing values in weather data.

4.2 Dimension Reduction in Weather Data

Weather data acquired from Daymet [32] consist of daily observations of six weather parameters listed in the data availability and dimensions, covering the years from 1980 to 2018. Totally 2190 trainable parameters for the proposed model to be trained on the basis of weather data alone, this leads to enormous weather trainable parameters. This was significantly reduced by the way of conversion of daily observation of weather data into weekly observations by the average method leads to dimension reduction in the ratio of 365:52. Weather trainable parameters were significantly reduced from 2190 to 312 for training our model and got better prediction than daily observations of weather data.

4.3 Lasso Based Feature Selection (LFS)

Feature selection is the process of selection of most relevant features from the dataset for the enhancement of accuracy levels in prediction as well as classification problems. Corn and Soybean dataset consist of 434 and 432 features respectively. We used Lasso based feature selection technique and extracted 380 out of 434 features from the corn dataset. For soybean dataset, 360 out of 432 features were extracted as relevant features for crop yield prediction on both the crops. It mainly extracted the most relevant features based on the importance weights of all the features.

4.4 Correlation Based Feature Selection (CFS)

Correlation based feature selection is one among the effective techniques for the extraction of most correlated features. Many data scientists have used this technique for feature extraction in their machine learning algorithms. In our corn and soybean bean data set, the predicted yield feature is correlated with the features such as weather, soil and management features. There are many popular types of correlation used for feature selection. We used Pearson correlation for our regression model in order to extract mostly correlated features corresponding to yield. Pearson's correlation coefficient is the measure of division of covariance of variables two variables and the product of their standard deviations, mathematical representation of Pearson correlation is denoted as follows

$$\rho_{A,B} = \frac{cov(A,B)}{\sigma_A \sigma_B} \tag{1}$$

where $\rho_{A,B}$ is the Person correlation coefficient between two features A and B. $cov(A, B)$ is the covariance if two features A and B. $\sigma_A \sigma_B$ is the standard deviations of the features A and B.

In the correlation based feature selection, crop yield data set has been divided into training testing parts, after some data pre-processing for feature engineering in terms of filling missed values was carried out, then calculation of highly correlated feature was done. As a result

dimensionality reduction or most correlated features extraction from initial dataset and finally the extracted features are fed into suitable machine learning algorithms for crop yield prediction as clearly shown in the Fig. 1.

After the extraction of 380 and 360 relevant features based on lasso feature selection technique. We attempt to extract the features which are highly correlated towards crop yield by using this correlation based feature selection technique. As a result nearly 25 features are omitted as they were no way correlated with crop yield. It includes 21 weather parameters and 4 crop management features were not correlated with crop yield. We extracted the remaining correlated features of corn and soybean crop with respect to crop yield with their appropriate scores. Based on the scores of correlated features, around 170 features of corn and 192 features of soybean are correlated with crop yield.

4.5 Mutual Information Based Feature Selection (MIFS)

Mutual information based feature selection is the power statistical method to identify the feature relationship between the datasets. It is also termed as the elimination process of reducing the input size of the dataset by extracting the mutual relationship between the features without affecting the most relevant features supporting classification or regression problem. Feature selection can be carried out through finding the dependence between the random features should be always symmetric and non-negative. It should select only the independent features which have an indication of zero.

The Mutual information between two discrete random features $A=(a_1, a_2, \ldots, a_k)$ and $B=(b_1, b_2, \ldots, b_k)$ is defined as

$$I(A,B) = \sum_a \sum_b p(a,b) \log \frac{p(a,b)}{p(a)p(b)} \tag{2}$$

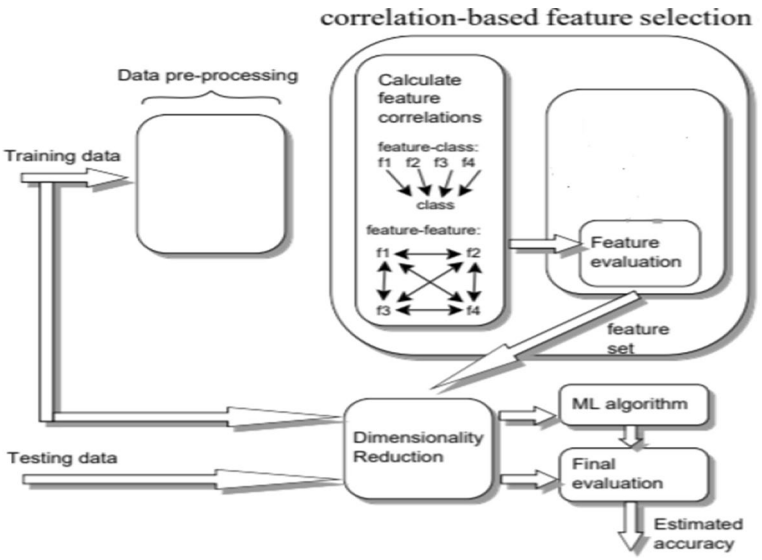


Fig. 1 Correlation-based feature extraction

where $I(A, B)$ is the mutual information between two discrete random features A and B. (a_1, a_2, …, a_k) and (b_1, b_2, …, b_k) represent the values of the discrete features A and B. $p(a, b)$ represent joint density function. $p(a)$ and $p(b)$ represent marginal density functions.

In the Mutual information based feature selection, after some data preprocessing for feature engineering in terms of filling missed values was carried out from the training and testing data of initial dataset and then calculation of mutual information feature was done. As a result dimensionality reduction from the initial dataset and finally the extracted features are fed into suitable machine learning algorithms for crop yield prediction as clearly shown in the Fig. 2.

We attempt to identify highly independent features from mutual information features towards crop yield by using this mutual information based feature selection technique. As a result nearly 15 features are omitted as they were no way related with crop yield. It includes 15 weather parameters that are not influenced with crop yield. We extracted the remaining independent features of corn and soybean crop with respect to crop yield with their appropriate scores. Based on the scores of mutual information features, around 270 features of corn and 290 features of soybean are correlated with crop yield.

5 Models Analysed

Regression techniques are widely used in prediction problems in machine learning. Our proposed model also engaged with regression techniques in order to predict the crop yield affected with multiple environmental factors such as weather, soil and plant management parameters. Various regression techniques are compared with our proposed model in terms of accuracy levels in yield prediction with various accuracy metrics such as RMSE, MAE and MSE.

5.1 Multiple Linear Regression (MLR)

Multiple linear regression is used to predict the output variable based on the multiple input variables. In machine learning MLR is a statistical model to find out the linear

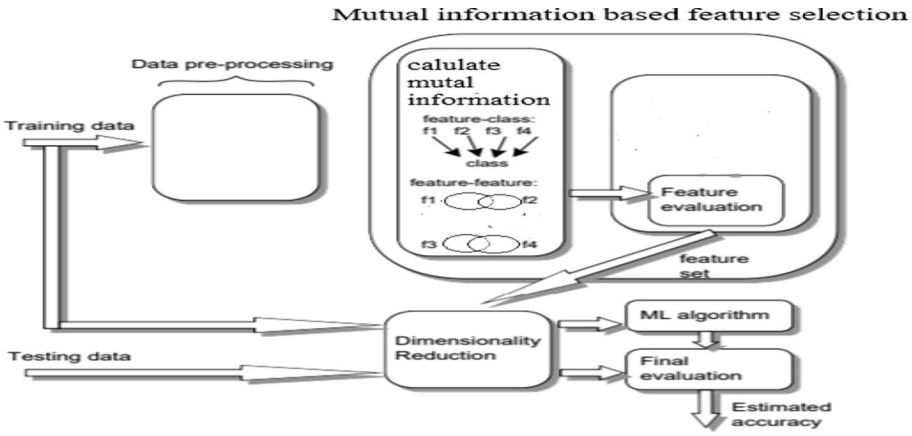


Fig. 2 Mutual information based feature selection

relationship between dependent or targeted output variable and independent input variable. MLR more suits our problem since the target yield variable has a linear relationship with other independent input variables such as weather, soil and other variables. Applying mathematical notation

$$Y_i = C_0 + C_1 w_{i1} + C_2 w_{i2} + \cdots + C_p w_{ip} + \varepsilon \tag{3}$$

where Y_i termed as output or dependent variable, w_i is input variable or independent variables, C_0 is Constant, C_p is coefficients for input variables and ε is error.

5.2 Decision Tree Regression (DTR)

Decision tree regression model is used to make decisions for predicting the output or targeted variable using the single decision tree. Dividing the overall observations of a particular process into a smaller subset of observations, decisions are made from the smaller observations are integrated leads to the process of final decision or final prediction using decision tree as shown in the Fig. 3. Decision tree regression also suits crop yield prediction since yield prediction can be made with number of decisions based on environmental and other factor of plant by using a Decision tree.

5.3 Random Forest Tree Regression (RFTR)

Random forest tree regression is involved in the process of predicting the output or targeted variable from various input or independent variables with the help of multiple decision trees for prediction. The Decisions or output from various decision trees are taken into consideration and the final prediction is based on averaging all the results obtained from various decision trees as shown in the Fig. 4. This model is also applied to our research problem, by predicting the crop yield with multiple decision trees based on various crop yield affecting factors.

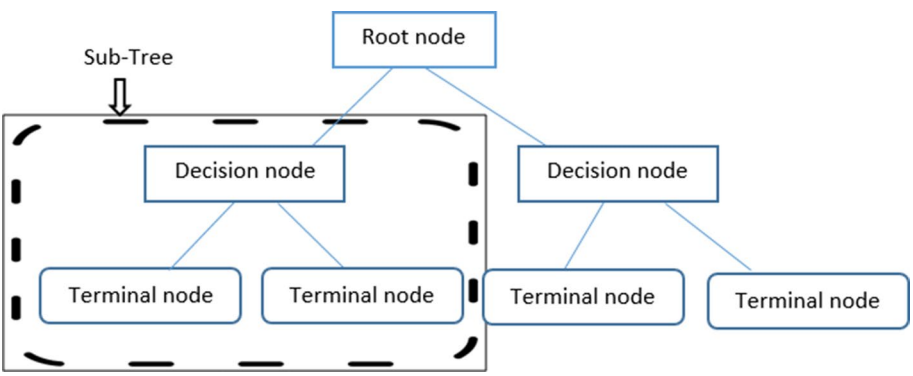


Fig. 3 Decision tree

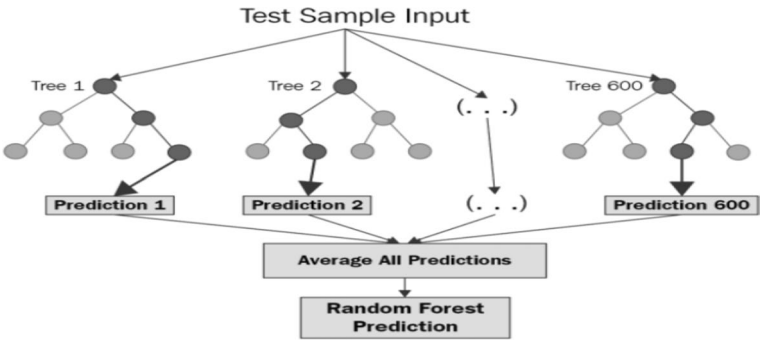


Fig. 4 Random forest tree regression

5.4 Gradient Boosting Regression (GBR)

Gradient boosting regression is an ensemble machine learning technique used to predict the target variable from a sequence of the learning models. Final prediction can be made on the basis of sequence learning from the previous regression models. It learns from the experience of sequences of previous regression models, so that it results in reduced bias, variance and higher accuracy rate in predictions as shown in Fig. 5. This technique also applied in the crop yield prediction since the final crop yield prediction can be learnt sequentially from various regression models like decision trees. Learning process in the basis accuracy of prediction with various accuracy metrics like RMSE, MAE and MSE from the subsequent prediction models.

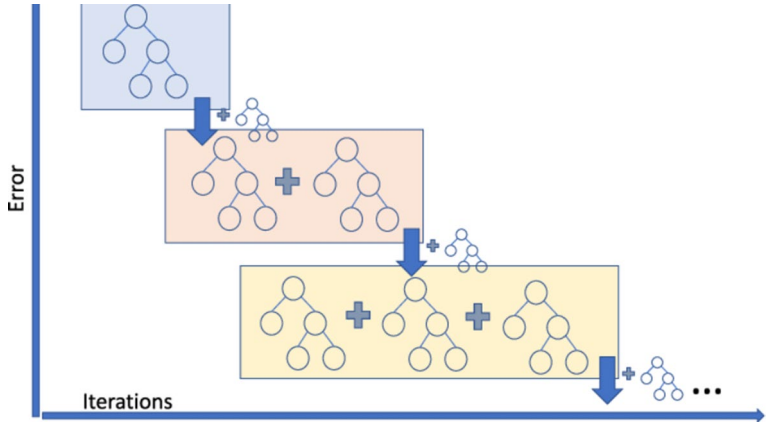


Fig. 5 Schematic representation of gradient boosting regression

6 Methodology and Metrics

Our proposed model, Mutual information feature selection (MIFS) based multilayer stacked ensemble regression (MSER) falls under the category of supervised machine learning involved in crop yield prediction or estimation.Since in the supervised machine learning process, the learning happened from the continuous observations on input or independent variables and output or target variables.Likewise our crop yield prediction model also lies on continuous learning from the year wise and locations based independent features like weather, soil and crop management features. Crop yield values will be acted as output or target variables. Regression and classification are the two well known types of supervised machine learning, our proposed crop yield prediction model lies under the category of regression type as it involved in the process of crop yield prediction category.Our proposed crop yield prediction models analysed with various feature selection techniques and crop yield accuracy results were compared with various machine learning regression algorithms.

The Data collection part consists of crop phenotype or environmental factors such as weather, soil, crop management and crop yield data were collected from various sources of the United States, covering 38 years of yield data along with 104 different locations. Proper feature Engineering and feature selection techniques are also involved in the process of feature selection or extraction, then modeling is done with proposed crop yield prediction model and final predicted crop yield values are analysed with various machine learning techniques as shown in Fig. 6.

6.1 MIFS Based Multilevel Stacked Ensemble Regression

Our proposed crop yield prediction model primarily focused on the prediction of crop yield production applying mutual information feature selection based multilevel stacked ensemble regression. Since mutual information based feature selection extracts the most related features towards crop yield when compared to the correlation based feature selection technique. Experimental analysis made with two feature selection techniques, mutual information based feature selection outperformed correlation based feature selection with

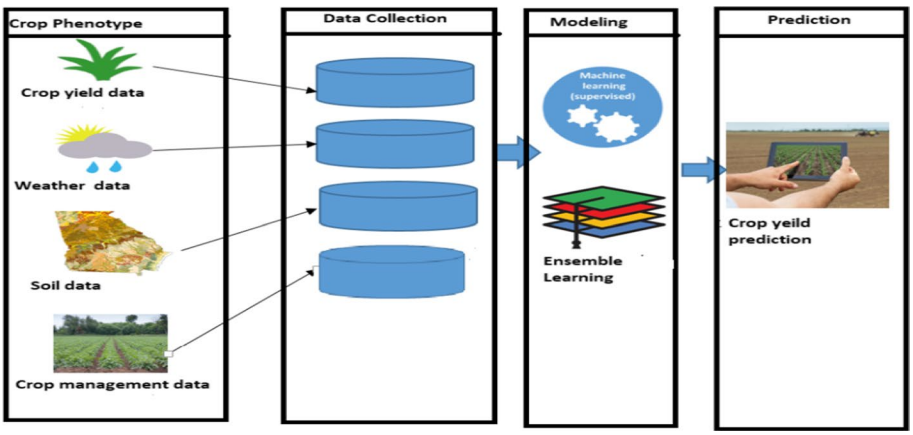


Fig. 6 Architecture diagram of crop yield prediction

270 features of corn crop and 290 features of soybean crop are most influential features produced higher yield accuracies with all other machine learning and ensemble learning models.

Initially the learning process of multilevel stacked ensemble regression model starts from its first level consist of one or more machine learning models, based on the learning outcome of first layer machine learning models in terms of error rate, prediction accuracy level and most influential features towards crop yield were experimentally analysed and comparisons were made with actual and predicted crop yield values, which in turn they are assumed as the weak learners. The second level of stacked ensemble regression also consists of one or more machine learning models generally termed as Meta-models. The outcomes of weak learners will act as the input for the meta-model. The learning process of meta-model completely based on the weak learners predictions observations, error rate and crop yield influential features. Complete analysis and comparisons made on weak learners, based on identification of non-influential features towards crop yield, identification of learning rate of first layer or weak learners machine learning model with respect to error rate are gathered as experiences from the weak learners. Final crop yield predictions of meta-model based on the experiences and results gathered from the weak learners and produced crop yield predictions on its own capabilities and experience gained from the weak learners. Multilevel stacked ensemble regression Multilevel Stacked ensemble regression consisting of multiple layers consist of weak learners followed by the sequence layers of meta-model. The Final layer of meta-model combines all the results or experience gathered from other weak learner and meta-models involved in the process of crop yield predictions as shown in the Fig. 7.

Our Mutual information feature selection based Multi-layer stacked ensemble regression model performed well when compared to other bagging and boosting ensembles. In boosting and bagging ensembles, the learning process carried out through the combination of homogenous machine learning techniques most commonly as decision trees. The multi-level stacked ensemble regression, the learning process was carried out through the combination of heterogeneous week learners. If any homogeneous learning model doesn't fit the crop yield dataset and does not produce higher accuracy, it may be reflected in the final prediction of crop yield.

Multilayer stacked ensemble regression predictions based on the sequential learning process form the weak learners. In the first layer consist of some of the supervised machine learning algorithms namely K-nearest neighbour, decision tree regression,

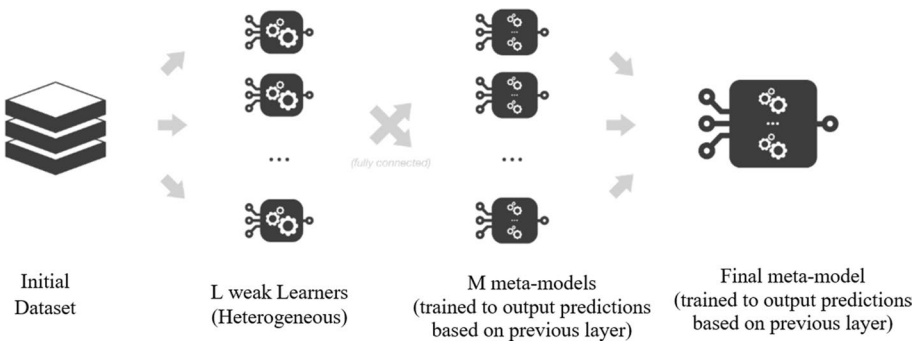


Fig. 7 Multilevel stacked ensemble regression

support vector machine were treated as weak learners as they are engaged in prediction of crop yield. Multiple linear regression and Random forest regression were treated as Meta-model, which could learn based on the previous results of weak learners. In the final layer of meta-model, Gradient Boosting regression engaged for the final prediction of crop yield based on all the previous experience or prediction results of other Meta-models as well as weak learners.

Algorithm- MIFS based Multilevel stacked ensemble regression

Given: Set of observation $A = \{a_i \in R^M\}$, Prediction set $B = \{b_i \in N\}$, Training data set $D = \{(a_i, b_i)\}$

Input: $D = \{(a_i, b_i) \mid a_i \in A, b_i \in B\}$

Output: An Ensemble prediction P

1. **Step 1**: Learn first-level predictions
2. For $t \leftarrow 1\ to\ T \quad do$
3. Learn a base predictions $p_t\ based\ on\ D$
4. **Step 2**: Construct new dataset from D
5. For $i \leftarrow 1\ to\ m \quad do$
6. Construct a newly extracted data set contains $\{a_i^{new}, b_i\}$ where $a_i^{new} = \{p_j(a_i) for\ j = 1\ to\ T$
7. **Step 3**: Learn second level predictions
8. For $t \leftarrow 1\ to\ T \quad do$
9. Learn a base predictions $p_t\ based\ on\ D$
10. Construct a newly extracted data set contains $\{a_i^{new}, b_i\}$ where $a_i^{new} = \{p_j(a_i) for\ j = 1\ to\ T$
11. **Step 4:** Learn third level predictions
12. Learn a new predictions p^{new} based on the newly extracted dataset.
13. **Return** $P(x) = p^{new}(p_1(a), p_2(a) \ldots, p_T(x))$

Crop yield performance dataset of corn and soybean crops consist of phenotype factors such as weather, soil and crop management data, our mutual information multilevel stacked regression involved in the process of crop yield prediction. Data pre-processing includes conversion of Daily recorded weather observation into weekly observations with the help of cumulative mean of week days and some missed values in soil and crop management data are replaced with same type of variables available in other counties. After preprocessing for the feature engineering process, the feature selection process was carried out through mutual information based feature selection. Division of training and testing from crop yield dataset, our crop yield prediction model based on multilayer stacked ensemble regression along with other ensemble and machine learning algorithms are compared with the yield results with the

regression evaluation parameter metrics as shown in Fig. 8 represents the complete workflow of our model.

6.2 Regression Accuracy Parameter Metrics

In the Regression problem, there are a number of metrics that were engaged in measuring the performance of the machine learning regression model in terms of prediction accuracy and error rate. They are listed as follows.

6.2.1 Percent Error

This is generally termed as measure of error percentage by the difference in the predicted and the original value divided by the original value.

$$\text{Percent Error} = \frac{predicted\ value - original\ value}{original\ value} \tag{4}$$

6.2.2 Mean Absolute Error (MAE)

MAE is generally termed as the measure of summation of absolute difference in the original and predicted values divided by the total number of observations.

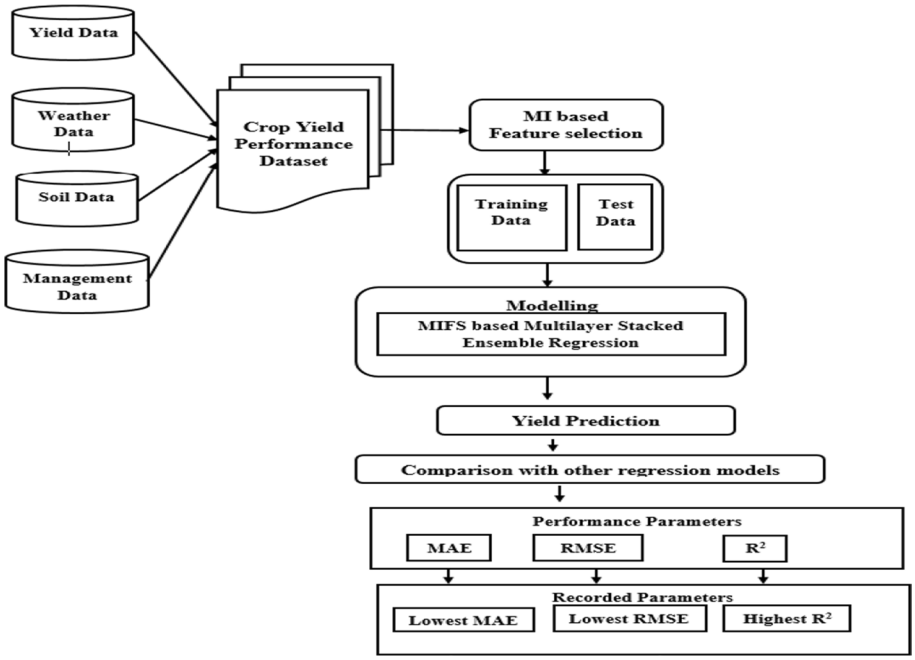


Fig. 8 Flow diagram of crop yield prediction

$$MAE = \frac{1}{N}\sum_{j=1}^{n}\left|x_j - \hat{x}\right| \tag{5}$$

where N is the total number of observations or data points in the dataset. x_j is the original value of x. $\hat{x}$ is the predicted value of x. $\sum_{j=1}^{n}\left|x_j - \hat{x}\right|$ is the summation of absolute difference in the actual and predicted values ranges from $j=1$ to n.

6.2.3 Mean Square Error (MSE)

MSE is generally termed as the measure of squared summation of absolute difference in the original and predicted values divided by the total number of observations.

$$MSE = \frac{1}{N}\sum_{j=1}^{n}\left(x_j - \hat{x}\right)^2 \tag{6}$$

where N is the total number of observations or data points in the dataset. x_j is the original value of x. $\hat{x}$ is the predicted value of x. $\sum_{j=1}^{n}\left(x_j - \hat{x}\right)^2$ is the squared summation of absolute difference in the actual and predicted values ranges from $j=1$ to n.

6.2.4 Root Mean Square Error (RMSE)

RMSE is generally termed as the square root of Mean Square Error.

$$RMSE = \sqrt{MSE} \quad \text{or} \quad \sqrt{\frac{1}{N}\sum_{j=1}^{n}\left(x_j - \hat{x}\right)^2} \tag{7}$$

where $\sqrt{}$ is the square root. N is the total number of observations or data points in the dataset. x_j is the original value of x. $\hat{x}$ is the predicted value of x. $\sum_{j=1}^{n}\left(x_j - \hat{x}\right)^2$ is the squared summation of absolute difference in the actual and predicted values ranges from $j=1$ to n.

6.2.5 R-Squared

R-squared is also termed as Coefficient of determination. It represents the closest determination of coefficients to the original values. R-Squared values lie between 0 and 1 where 0 represents the model doesn't fit with the dataset, 1 represents the model perfectly fit with the given data.

$$R^2 = 1 - \frac{\sum\left(x_j - \hat{x}\right)^2}{\sum\left(x_j - \bar{x}\right)^2} \tag{8}$$

where x_j is the original value of x. $\hat{x}$ is the predicted value of x, $\bar{x}$ is the mean value of x. N is the total number of observations or data points in the dataset, $\sum\left(x_j - \hat{x}\right)^2$ is the square of the difference in the actual and predicted values. $\sum\left(x_j - \bar{x}\right)^2$ is the square of the difference in the actual values and mean value of x.

7 Results and Discussion

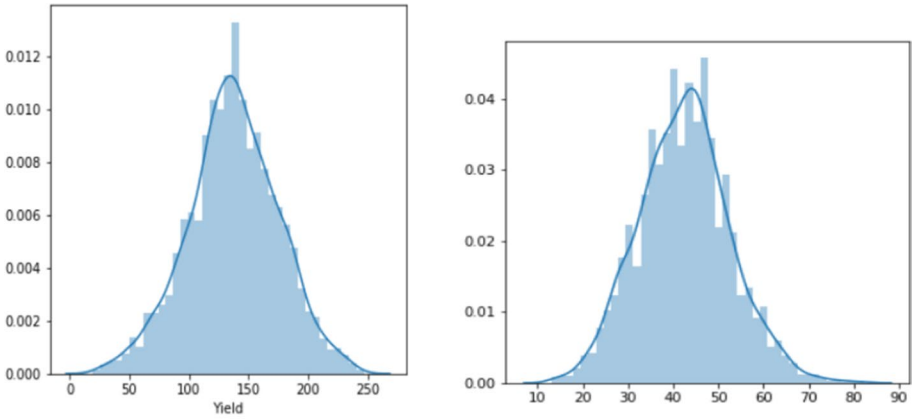
Using python packages of Tensor flow as the rear end and keras on the top, our proposed mutual information based multilayer stacked ensemble regression crop yield prediction model was trained and tested. Crop related data collected from various sources includes weather, soil and management factors of crop across major counties of the United States from the duration between 1980 and 2018. As we already listed six weather components recorded on the daily basis covering 38 years ranges between 1980 and 2018 across 12 counties, in the data availability section. Daily observations of weather data for the duration of 38 years leads to drastic increase in the weather parameters belonging to independent variables. This results in dimensionality reduction on weather data by the conversion of daily observations into weekly observations as in the ratio 365:52 and it leads to significant decrease in trainable parameters and good fit for learning. There are 10 soil components taken into consideration for soil data collected across 12 states, 38 years of duration. It also consists of 16 and 14 components respectively for corn and soybean crops throughout the stipulated duration and locations. The measurement unit of crop yield can be represented as bushels per acre will be represented as output or target variable. The input or independent variables are represented as weather, soil and crop management related data. Totally 434 and 432 trainable or independent features used to train our crop prediction model for corn and soybean crops respectively. Overall count of 1,740,435 and 1,732,433 records are found in the corn and soybean crop yield dataset. Table 2 shows the detailed representation and values of weather, soil and management data in the corn and soybean crop dataset. Figure 9a, b demonstrates the distribution of corn and soybean crop yield data points based on the various input or independent variables. Figure 10a, b demonstrates the distribution of year, yield and few weather components based on it values across the dataset.

7.1 Evaluation of CFS and MIFS

CFS technique also tried for our crop yield prediction model to extract most correlated features with respect to the crop yield by the Pearson's correlation measure. The scikit-learn library provides the function termed as f_regression() used for the extraction of correlated features and selection of top k correlated features are extracted through SelectKBest class. By using these around 170 and 192 mostly correlated features of corn and soybean crops

Table 2 Conceptual representation of crop yield performance dataset

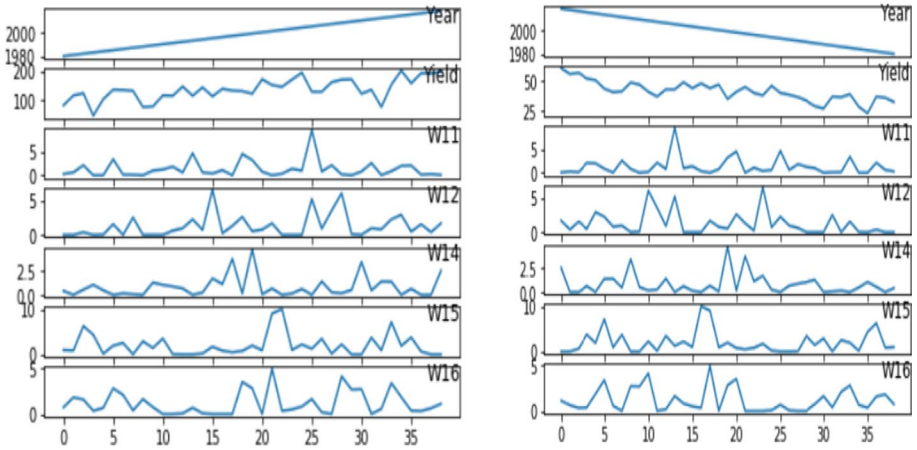
Loc_id	Year	Yield	W11	W12	W13	…	P11	P12	P13	P14	P15	P16	SS1	SS2
0	2018	190.3	0.011	1.059	0.821	…	37	15	9	9	4	2	1.81	0.81
0	2017	204.1	0.035	1.296	6.559	…	9	25	14	14	23	8	1.81	0.81
0	2016	215.6	0.042	2.214	0.142	…	10	22	21	21	9	9	1.81	0.81
0	2015	182.9	3.071	0.678	0.119	…	21	14	22	22	13	6	1.81	0.81
0	2014	194.1	2.250	2.821	0.880	…	23	10	28	28	21	7	1.81	0.81
…	…	…	…	…	…	…	…	…	…	…	…	…	1.81	0.81
104	1990	129.6	0.742	0.171	4.685	…	23	36	14	14	4	1	1.81	0.81
104	1989	132.0	2.857	4.257	1.426	…	14	31	12	12	19	14	1.81	0.81
104	1988	79.1	0.142	0.142	5.028	…	9	5	26	26	9	9	1.81	0.81
104	1987	135.2	0.285	0.285	2.142	…	0	1	11	11	22	25	1.81	0.81



a *Yield Distribution summary of corn crop*

b *Yield Distribution summary of soybean crop*

Fig. 9 **a** Yield distribution summary of corn crop. **b** Yield distribution summary of soybean crop



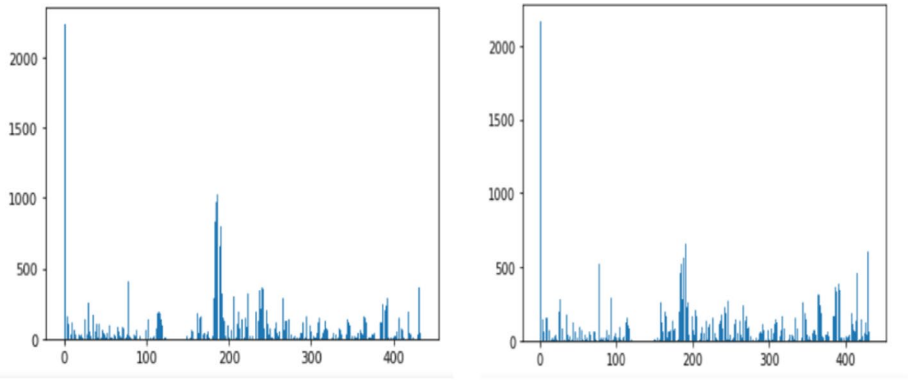
a *Distribution of Year, yield and weather data in corn crop*

b *Distribution of Year, yield and weather data in soybean crop*

Fig. 10 **a** Distribution of year, yield and weather data in corn crop. **b** Distribution of year, yield and weather data in soybean crop

were selected based on the scores of every feature. Figure 11a, b shows the correlation based feature selection of corn and soybean crops on scores. Table 3(a) and (b) shows the top 10 features selection based on correlation based feature selection.

MIFS technique also applied to our crop yield prediction model to find the mutual relationship between the features and independent features are selected respect to the crop yield. Here also scikit-learn library provides the function termed as f_regression() used for the extraction of mutual information based features and selection of top



a *CFS based feature selection of corn crop*

b *CFS based feature selection of Soybean crop*

Fig. 11 **a** CFS based feature selection of corn crop. **b** CFS based feature selection of soybean crop

Table 3 Top 10 feature selection of (a) corn crop based on CFS, (b) soybean crop based on CFS

Features	Scores
(a) Corn crop based on CFS	
Year	0.66652
P6	0.49806
P7	0.43693
P8	0.41463
P5	0.40564
P9	0.4046
P11	0.28637
P10	0.28443
W431	0.25177
W428	0.2479
(b) Soybean crop based on CFS	
Year	2169.9117
W433	785.07825
W434	663.19393
S_surface2	609.36172
W431	566.56434
S_surface3	559.1973
W432	553.53116
W225	543.01663
W428	520.67158
W224	516.70014

K features are extracted through SelectKBest class. Around 270 and 290 mutual information based independent features of corn and soybean crop were selected based on the scores of every features. Figure 12a, b shows the mutual information based feature selection of corn and soybean crops on scores. Table 4(a) and (b) shows the top 10

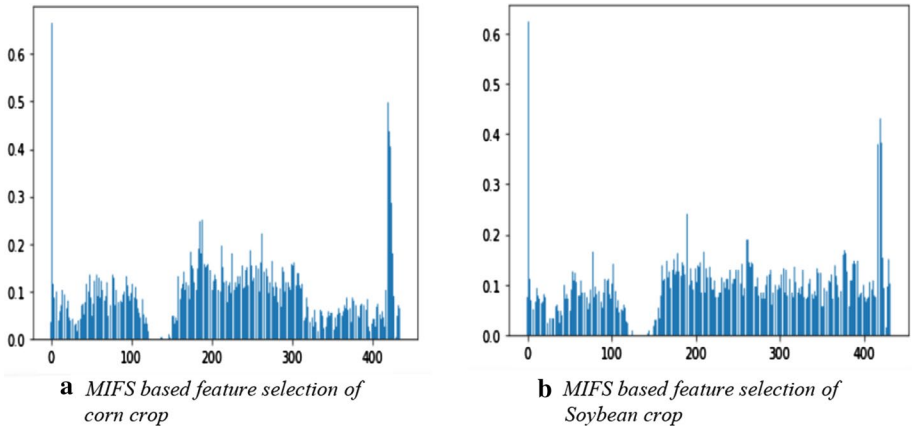


Fig. 12 **a** MIFS based feature selection of corn crop. **b** MIFS based feature selection of soybean crop

Table 4 Top 10 feature selection of (a) corn based on MIFS, (b) soybean based on MIFS

Features	Score
(a) Corn based on MIFS	
Year	2238.8032
W429	1021.8916
W428	968.75058
W431	922.47905
W426	828.84666
W433	796.7278
W427	754.44752
W430	662.93253
W432	655.47983
W224	550.39321
(b) Soybean based on MIFS	
Year	0.629509
P5	0.479387
P7	0.431878
P3	0.409615
P9	0.40033
P8	0.392411
P4	0.383969
P6	0.37198
P2	0.328144
W433	0.237379

features selection based on mutual information based feature selection. Our crop yield prediction model gives better performance in terms of yield prediction based on MIFS than CFS as shown in the Table 5(a) and (b).

Table 5 Actual versus predicted values of multiple linear regression in (a) corn crop, (b) soybean crop

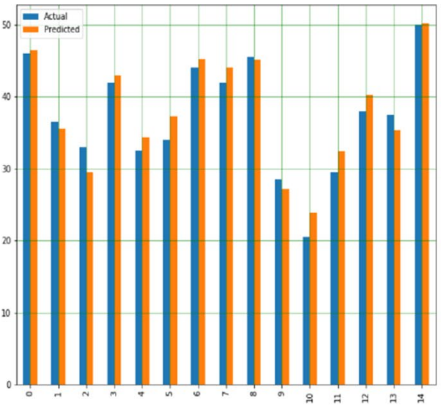
Actual yield	Predicted yield
(a) Corn crop	
141.0	148.426
105.0	107.762
41.0	48.038
117.0	123.131
115.0	119.133
162.2	162.626
169.0	156.944
85.0	75.782
164.0	157.234
134.0	130.653
110.0	111.788
120.0	112.981
114	119.230
(b) Soybean crop	
46.0	46.4555
36.5	35.551
33.0	29.457
42.0	42.947
32.5	34.283
34.0	37.195
44.0	45.175
42.0	44.007
45.5	45.072
28.5	27.192
20.5	23.937
29.5	32.385
38.0	40.224

7.2 Comparative Analysis of MIFS based Multi-layer Stacked Ensemble Regression

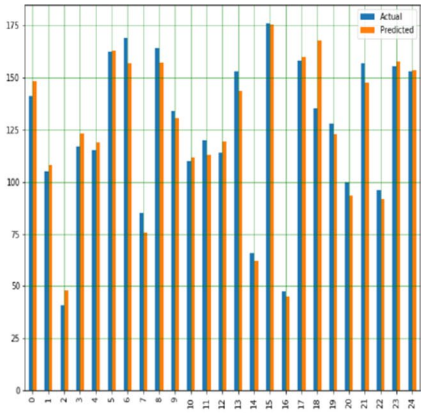
A complete comparative analysis are made between our proposed crop yield prediction model based on Multi-layer stacked ensemble regression and other machine learning models. Yield accuracy percentage, regression accuracy parameter metrics, measurement between actual and predicted crop yield values were considered as predominant factors for comparative analysis.

MLR involved in our crop yield prediction model for analysis, since multiple environmental, soil and management parameters included in the process of crop yield prediction. Using Sklearn linear regression library provides LinearRegression() in which the training and testing data will be fitted and prediction by using fit() and predict() method. Figure 5a, b shows the actual and predicted crop yield values of corn and soybean crops using MLR. Figure 13a, b shows graphical representation of actual and predicted crop yield values of corn and soybean crop using MLR.

Analysis of crop yield prediction is also done with DTR, by splitting of decision making features or nodes in the form decision tree and final crop yield prediction based on



a *Comparison between actual and predicted crop yield of MLR in corn crop*



b *Comparison between actual and predicted crop yield of MLR in soybean crop*

Fig. 13 **a** Comparison between actual and predicted crop yield of MLR in corn crop. **b** Comparison between actual and predicted crop yield of MLR in soybean crop

the decision nodes. The Sklearn tree regression library provides DecisionTreeRegressor() using maximum depth of 6, in which the training and testing data will be fitted and prediction by using fit() and predict() methods. Table 6(a) and (b) shows the actual and predicted crop yield values of corn and soybean crops using DTR. Figure 18a, b shows graphical representation of actual and predicted crop yield values of corn and soybean crop using DTR.

RFTR also involved in the analysis of crop yield prediction also done with DTR, by splitting of decision making features into number of decision trees and final crop yield prediction based on the combined decision of decision tree aggregation. RFTR falls under the category of advanced ensemble algorithms termed as bagging. The Sklearn tree regression library provides RandomForestRegressor () using maximum depth of 10, assigning a value 110 to the nestimators in which the training and testing data will be fitted and prediction by using fit() and predict() methods. Table 7(a) and (b) shows the actual and predicted crop yield values of corn and soybean crops using RFTR. Figures 14a, b and 15a, b shows graphical representation of actual and predicted crop yield values of corn and soybean crop using RFTR.

Analysis of crop yield prediction is also done with GBR, by the sequence learning from the other regression models in order to reduce bias, variance and improvement in prediction accuracy. Most commonly GBR learning processes should be carried out through homogeneous regression models, DT are commonly used. The Sklearn tree regression library provides GradientBoostingRegressor() using maximum depth of 4, assigning a value 500 to the nestimators in which the training and testing data will be fitted and prediction by using fit() and predict() methods. Table 8(a) and (b) shows the actual and predicted crop yield values of corn and soybean crops using GBR. Figure 16a, b shows graphical representation of actual and predicted crop yield values of corn and soybean crop using GBR.

MIFS based Multi-layer stacked ensemble regression also involved in the process of crop yield prediction based on level based learning of heterogeneous weak learners and meta-learners. Final crop yield prediction is the combined prediction of heterogeneous regression models. Using Sklearn ensemble regression library provides StackingRegressor() having the parameters states starting level number and ending level number as estimators and final

Table 6 Actual versus predicted values of decision tree regression in (a) corn crop, (b) soybean crop

Actual yield	Predicted yield
(a) Corn crop	
141.0	139.963
105.0	114.406
41.0	50.740
117.0	133.009
115.0	136.718
162.2	183.476
169.0	139.963
85.0	50.571
164.0	153.791
134.0	136.820
110.0	116.360
120.0	128.051
114.0	114.403
153.0	154.150
(b) Soybean crop	
46.0	45.137
36.5	34.776
33.0	32.233
42.0	44.504
32.5	34.776
34.0	34.894
44.0	44.504
42.0	44.504
45.5	44.504
28.5	27.342
20.5	23.810
29.5	35.443
38.0	35.443
37.5	33.838

estimators, with appropriate cross validation, in which the training and testing data will be fitted and prediction by using fit() and predict() method. Table 9(a) and (b) shows the actual and predicted crop yield values of corn and soybean crops using MIFS based Multi-layer stacked ensemble regression. Figure 17 shows the performance comparison of Multi-layer stacked ensemble regression with other machine learning models.

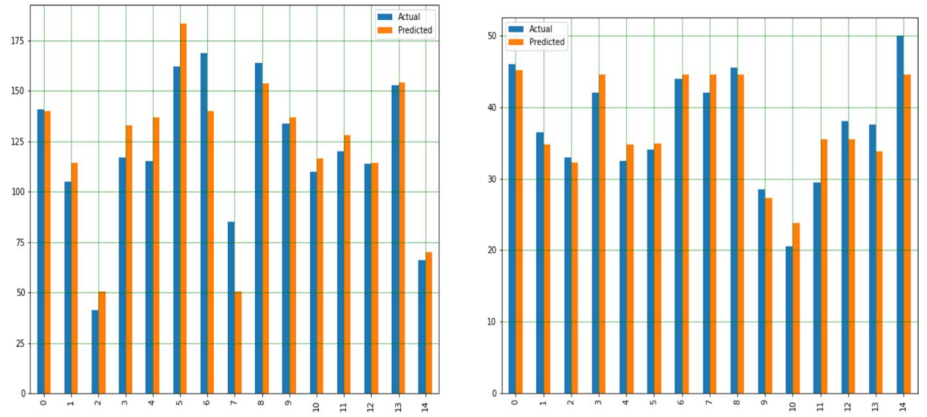
The Table 10(a) and (b) shows that the prediction accuracy comparison between the MIFS based Multi-layer Stacked Ensemble and other machine learning models used for crop yield prediction with evaluation of regression error metrics. Figure 18 shows the graphical representation of comparison of accuracies from different prediction models.

Table 7 Actual versus predicted values of random forest regression in (a) corn crop, (b) soybean crop

Actual yield	Predicted yield
(a) Corn crop	
141.0	138.563
105.0	110.406
41.0	45.740
117.0	129.432
115.0	130.758
162.2	181.476
169.0	145.963
85.0	65.571
164.0	158.791
134.0	135.860
110.0	114.860
120.0	124.051
114.0	115.403
153.0	154.150
(b) Soybean crop	
46.0	47.912
36.5	37.263
33.0	29.959
42.0	41.876
32.5	35.938
34.0	35.487
44.0	44.168
42.0	43.731
45.5	45.353
28.5	28.748
20.5	23.370
29.5	33.629
38.0	36.490
37.5	35.001

8 Conclusion

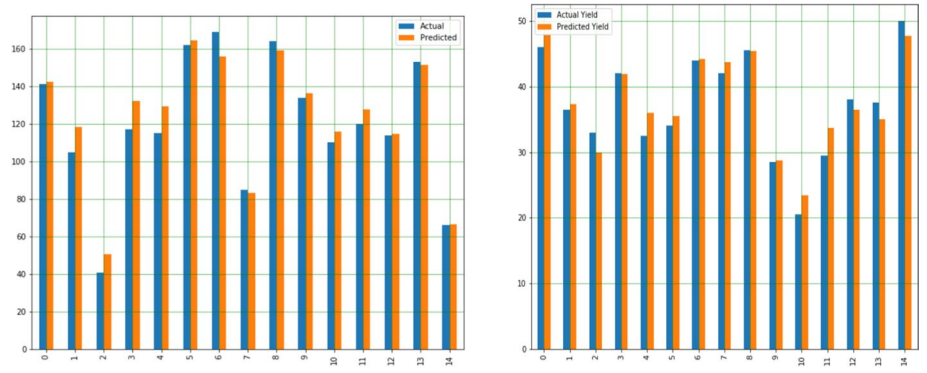
In global food production, crop yield estimation or prediction plays a vital role. Effective crop yield estimation provides great support for the decision makers to make suitable decisions about food products exports and imports to meet the national food demand. Due to drastic climatic changes, it affects the regular pattern in the rainfall, organic changes in the soil and other changes which significantly affects the farmers in terms of not getting the estimated crop yield. The crop yield performance dataset of corn and soybean collected from various sources and it comprises weather, soil and crop management parameters are treated as a feature set. Overall 434 features of corn and 432 features of soybean over the period of 38 years across 105 different locations of corn and soybean belt. MIFS technique involved in the feature selection process, 270 and 290 most related features of corn and soybean crops are extracted out of 434 and 432 features. MIFS extracts more number related features towards crop yield, whereas



a *Comparison between actual and predicted crop yield of DTR in soybean crop*

b *Comparison between actual and predicted crop yield of DTR in soybean crop*

Fig. 14 **a** Comparison between actual and predicted crop yield of DTR in soybean crop. **b** Comparison between actual and predicted crop yield of DTR in soybean crop



a *Comparison between actual and predicted crop yield of RFTR in corn crop*

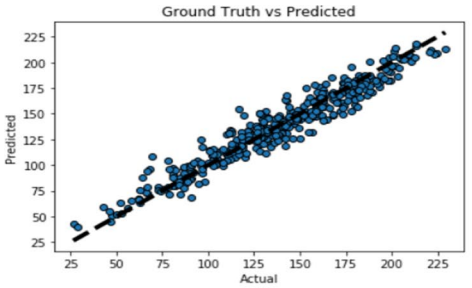
b *Comparison between actual and predicted crop yield of RFTR in soybean crop*

Fig. 15 **a** Comparison between actual and predicted crop yield of RFTR in corn crop. **b** Comparison between actual and predicted crop yield of RFTR in soybean crop

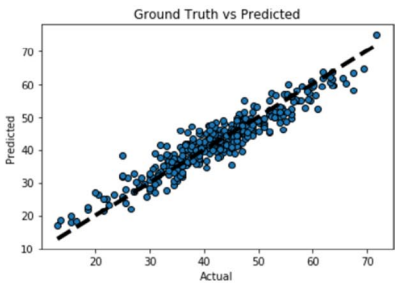
CFS extracts less number of correlated features which drastically reduces the prediction of yield accuracy performance. Our proposed MIFS based Multilayer stacked ensemble regression focused on prediction of crop yield not only based on single crop yield value factor but it also includes the other crop yield affecting factors such as weather, soil and crop management parameters. The prediction process of crop yield is based on the results and experiences of the weak learners in the form of a multi-layered approach. Our prediction results shows that weather and crop management features are most influential towards crop yield predictions rather than soil features. Comparative analysis also made with various machine learning regression such as MLR, DT, SVM and KNN, advanced ensemble learning algorithms such as RF and GBR also compared. Proposed

Table 8 Actual versus predicted values of GBR in (a) corn crop, (b) soybean crop

Actual yield	Predicted yield
(a) Corn crop	
97.0	118.187
163.0	144.974
102.0	108.996
80.6	91.370
91.0	86.806
132.0	123.370
159.0	147.563
101.0	114.975
119.5	148.434
133.0	134.800
142.0	144.680
(b) Soybean crop	
34.5	36.024
45.0	41.339
13.0	17.154
46.0	43.893
27.0	32.673
37.0	35.864
37.5	41.194
42.5	41.360
37.0	45.330
39.0	40.129
43.8	48.025



a *Comparison between Ground truth Vs Predicted crop yields of GBR*



b *Comparison between Ground truth Vs. Predicted crop yields of GBR*

Fig. 16 **a** Comparison between ground truth versus predicted crop yields of GBR. **b** Comparison between ground truth versus predicted crop yields of GBR

MIFS based Multilevel Stacked Ensemble Regression outperformed other machine learning model and other advanced ensemble learning algorithms with the prediction accuracy of 94.439% having lowest MAE and RMSE values of 6.63, 10.545 respectively

Table 9 Actual versus predicted values of MSER in (a) corn crop, (b) soybean crop

Actual yield	Predicted yield
(a) Corn crop	
162.0	157.524
174.0	179.181
161.0	158.219
174.3	184.917
110.0	110.241
133.0	134.800
142.0	144.680
146.0	147.745
179.0	178.058
145.0	146.856
(b) Soybean crop	
43.5	45.033
35.5	34.487
46.0	44.945
35.5	37.376
46.0	46.624
46.0	42.212
48.0	45.610
50.5	48.181
41.0	41.466
41.0	42.288

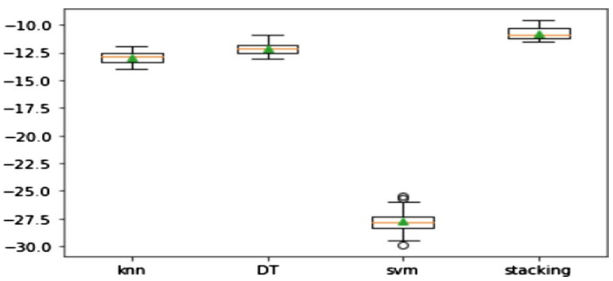


Fig. 17 Performance comparison of MSER

for the corn crop and prediction accuracy of 92.426% having lowest MAE and RMSE values of 7.431, 11.005 respectively for the soybean crop.

In future work, crop yield prediction models may be experimented with time series analysis in which estimations predominantly depend on the observations of yield pattern changes over the time. Addition of genotype features along existing phenotype features in the crop yield performance dataset and finding the most influential and non-influential genotype features towards better crop yield prediction is a research challenge that still exists.

Funding Not applicable.

Table 10 Performance comparison table of (a) corn crop, (b) soybean crop

Techniques	Yield accuracy (%)	MAE	MSE	RMSE
(a) Corn crop				
Multiple linear regression	88.388	10.117	139.425	11.807
Decision tree regression	81.698	12.601	270.215	16.438
Support vector machine regression	74.423	18.238	364.128	19.082
K-nearest neighbour regression	78.291	14.821	352.636	18.778
Random forest tree regression	90.848	8.68	135.121	11.624
Gradient boosting regression	92.648	7.39	120.231	10.964
MIFS based multilayer stacked ensemble	94.439	6.63	111.213	10.545
(b) Soybean crop				
Multiple linear regression	87.260	11.254	144.214	12.008
Decision tree regression	76.499	18.486	295.652	17.194
Support vector machine regression	72.521	20.432	402.154	20.253
K-nearest neighbour regression	77.841	13.453	360.976	18.976
Random forest tree regression	89.198	9.754	139.425	11.807
Gradient boosting regression	88.504	10.567	140.120	11.837
MIFS based multilayer stacked ensemble	92.426	7.431	121.123	11.005

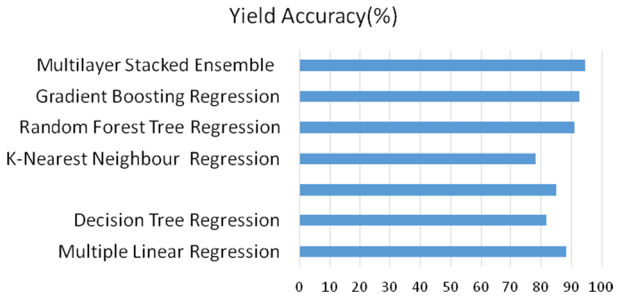


Fig. 18 Comparing the accuracies of different prediction models

Declaration

Conflict of interest We authors not having any conflict of interest among ourselves to submit and publish our articles in Wireless Personal Communications journal.

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